

JAYESH MOTWANI

motwanij63@gmail.com ♦ LinkedIn ♦ GitHub ♦ Portfolio ♦ 8817885445

EDUCATION

Bachelor of Technology, ECE, Vellore Institute of Technology
CGPA: 9.5

July 2024 – May 2028

EXPERIENCE

Product Engineering Intern (AI) | NewVision Software | On-Site

May 2026 – July 2026

- Developed AI agents using Azure AI Foundry and Foundry SDK to automate SDLC workflows, including User Story Generation, Impact Analysis, and Quality Assurance/Testing.
- Built retrieval-augmented agent pipelines leveraging Azure AI Search, custom indexing strategies, metadata enrichment, and semantic retrieval.
- Integrated AI agents with Azure DevOps Boards for automated user story analysis, finding defect hotspots, and workflow augmentation.
- Reduced production hallucinations through evaluation-driven development, prompt optimization, context management, state coherence, and retrieval quality improvements.
- Engineered scalable data ingestion pipelines and custom skillsets for document chunking, embedding generation, metadata extraction, and search indexing.
- Developed Azure Function Apps to address metadata parsing and preprocessing bottlenecks within enterprise document pipelines.
- Collaborated directly with clients and stakeholders to gather requirements, refine agent behavior, and deliver production-ready AI solutions.

RESEARCH

CV-Weighted Gradient Curriculum (CVGC)

- Novel Coeff. of Variation based decaying input weighting schedule to stabilize Adam on heterogeneous tabular data, derived formal complementarity theorems and proofs.
- **3,465 runs**, 11 datasets, 3 architectures, 10 seeds; **43–85% reduction** in validation loss area where Adam diverges.

Deep Generative Imputation for Residual Disease Prediction in Colorectal Cancer

- Designed a custom VAE with entity embeddings for mixed-type clinical data imputation and SMOTE-based class balancing in colorectal cancer biospecimen datasets.
- Achieved **AUC 0.98 ± 0.03**, significantly outperforming MICE (0.79) and Mean/Mode imputation (0.76).

Hybrid GNN for Ionic Liquid Viscosity Prediction

In Progress

- Developed a dual-stream GCN–MLP model integrating molecular graphs, 434 RDKit descriptors, and thermodynamic features for ionic liquid property prediction achieving **$R^2 = 0.886$** outperforming SOTA methods.

PROJECTS

Checkmate Machine | Python, OpenCV, CNN, Minimax

[Live](#) | [GitHub](#)

- Implemented real-time OpenCV pipeline for board detection and square segmentation and labeled a dataset for chess pieces classification using custom CNN at **97.5% accuracy**.
- Developed custom chess engine from scratch leveraging Minimax and Alpha-Beta pruning and board evaluation heuristics.

Privacy-First Legal RAG | Python, BERT, ChromaDB, sentence-transformers, Ollama

[GitHub](#)

- Engineered hierarchical PDF parser (Act → Section → Proviso), semantic search and conflict-detection across 25 legal conflicts. fine-tuned BERT NER masking **112+ PII types**.
- Offline inference via Ollama (qwen2.5:7b), zero external API calls.

SKILLS

AI Agents, Retrieval-Augmented Generation, Cloud AI Engineering, Agent Evaluation, Prompt Engineering, PyTorch, Scikit-Learn, XGBoost, Transformer Architectures, CNNs, Variational Autoencoders, Representation Learning, Time Series Forecasting, Azure AI Foundry, Azure Semantic Kernel, Azure AI Search, Azure Functions, Search Index Design, Metadata Enrichment Pipelines, Python, C++/C, SQL, NoSQL, FastAPI, SciPy, Docker, Git, ADO Board, OOP, Data Structures & Algorithms, Statistical Learning, Mathematics & Optimization, Feature Engineering, Model Explainability (SHAP)